



SMILING FOR THE CAMERA

During the first manned Apollo mission in October 1968, Walter Cunningham (right) holds up a sign for TV viewers on Earth, who were treated to a glimpse of the astronauts in orbit thanks to a camera inside the Apollo 7 module. The other crew members were Donn Eisele (left) and Wally Schirra.

stages, the first powered by five F-1 engines using a propellant mix of kerosene and liquid oxygen, and the other two with J-2 engines using a mix of liquid oxygen and liquid hydrogen – the propellant originally proposed by rocket pioneer Tsiolkovsky.

The three stages were joined together in the vast Vehicle Assembly Building at the Cape, on top of a platform with a 380ft (115m) high support tower. The rocket, platform, and tower were then all carried on a massive “crawler” to the launch pad three miles away. The first launch, on November 9, 1967, astounded everyone who witnessed it. The 7.5 million lb (3.4 million kg) of thrust created a shockwave like a minor earthquake – astronaut Mike Collins said it felt “as if a giant had grabbed my shirtfront and started shaking.” It was the heaviest object ever to fly, and the most powerful flying machine in history – and it worked.

In addition to continuing the Apollo Saturn missions, NASA was pursuing unmanned exploration of the Moon, partly as a preparation for a manned landing. Five Surveyor spacecraft set down on the Moon between May 1966 and January 1968, while five Lunar Orbiter spacecraft photographed almost the entire Moon surface between August 1966 and August 1967. These missions enabled NASA to identify sites where the lunar module would be able to land safely.

There was a prolonged interruption to manned flights while the problems in the command module revealed by the Apollo 1 fire were ironed

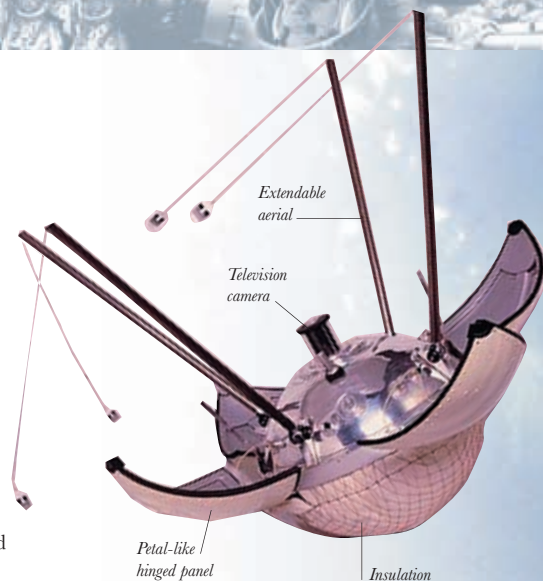
out. There were also delays in building the lunar module – not surprisingly as it contained around a million parts, each of which had to be as near perfection as humanly possible. And the second test launch of Saturn V, in April 1968, revealed serious glitches that fortunately were rapidly fixed. It was not until October 1968 that American astronauts returned to space, with Apollo 7 going into Earth orbit with Walter Schirra, Donn Eisele, and Walt Cunningham in the command module.

Soviet developments

A constant fear for NASA at this time was, of course, that the Soviets might leap in and put a man on the Moon with a one-time shot while the Apollo program was proceeding systematically toward its goal. In February 1966 the Soviet Union had landed the spacecraft Luna 9 on the Moon three months head of America's first Surveyor mission. It could be assumed that they would attempt similar coups to pre-empt the United States in the next two “firsts” – the first manned flight around the Moon and first manned Moon landing.

The Soviets were indeed planning for both types of mission, even though their space program had long ceased to compare with that of the United States in

scale and resources. The new unmanned Soyuz spacecraft, first put into orbit in 1966, was to be the basis for a circumlunar flight vehicle, the Zond, which would be launched by a three-stage Proton rocket. Soyuz was also to be the basis for a spacecraft to ferry two cosmonauts into lunar orbit, from where one of them would descend in a lunar lander to the surface of the Moon. This mission would require a more powerful launcher, the four-stage N-1. The chief difference between the Soviet plan for a manned Moon landing and its American equivalent was that the cosmonaut



LUNA 9

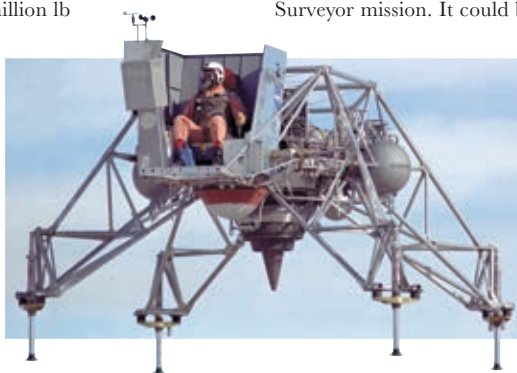
The first successful landing on the surface of the Moon was made by the unmanned Luna 9 spacecraft in February 1966. It was an achievement that masked the growing problems besetting the Soviet space program.

would have to perform a spacewalk to transfer back and forth between the main spacecraft and the lunar lander.

The Soviet government did not authorize a manned Moon program until 1964, and having started later than the Americans the Soviets soon fell even further behind. Initial unmanned test flights of Soyuz revealed problems that should have been solved before a manned flight was attempted. They were not. In April 1967, 40-year-old cosmonaut Vladimir Komarov, who had survived the incredibly risky three-man spacesuit-free Voskhod mission in October 1964, was sent up into orbit on board the bug-ridden Soyuz. He was supposed to rendezvous and dock with another Soyuz craft, but the second launch was quickly abandoned as Komarov's problems mounted. After only a day in space he was ordered to return to Earth. Partially out of control, Komarov had great difficulty in achieving reentry. When he did, the parachute on his spacecraft failed to deploy and he plunged into the ground at high speed. Komarov was the first person to die carrying out a spaceflight.

Race to orbit the Moon

After Komarov's death the Soviets carried out no further manned spaceflights for 18 months – a break that roughly paralleled the interruption of American manned missions after the Apollo 1 fire. A cosmonaut returned to space a week after the Apollo 7 mission, in October 1968, a manned Soyuz craft maneuvering close to an unmanned



LUNAR SIMULATOR

Lunar Landing Research Vehicles (LLRV) were designed to simulate the descent of a lunar module onto the Moon. Like giant flying headboards, they could be piloted down from around 1,500ft (500m) to the ground using the thrust from a jet engine. In 1967 they were converted into trainers for the astronauts.